Pelvic Viscera II

JeffMD 2026

EUR Block

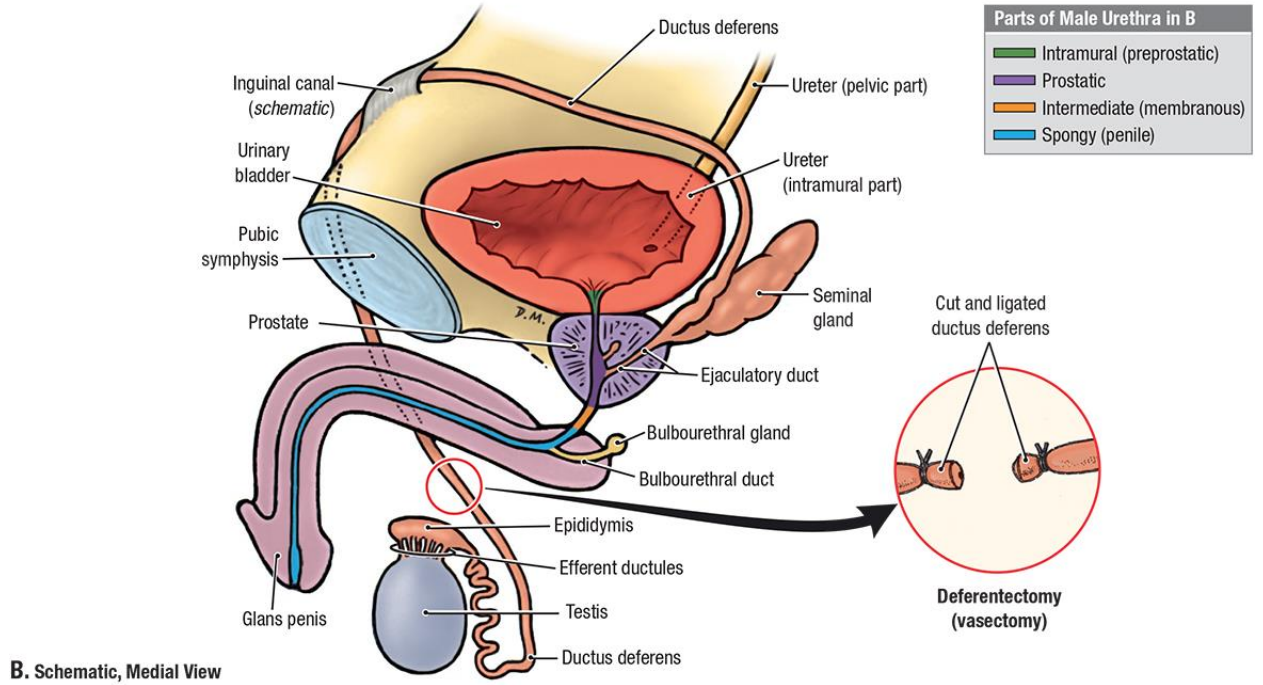
Elizabeth A Spudich, PhD

Learning Objectives Viscera II

- Describe the functional anatomy of the reproductive viscera concerned with Sperm maturation and transmission, including their anatomical relationships to abdominopelvic and perineal structures
- Describe the functional anatomy of the reproductive viscera concerned with Oocyte maturation and transmission, including their anatomical relationships to abdominopelvic and perineal structures
- Describe the sensory innervation of the uterus and vagina and how it impacts administration of anesthesia during childbirth

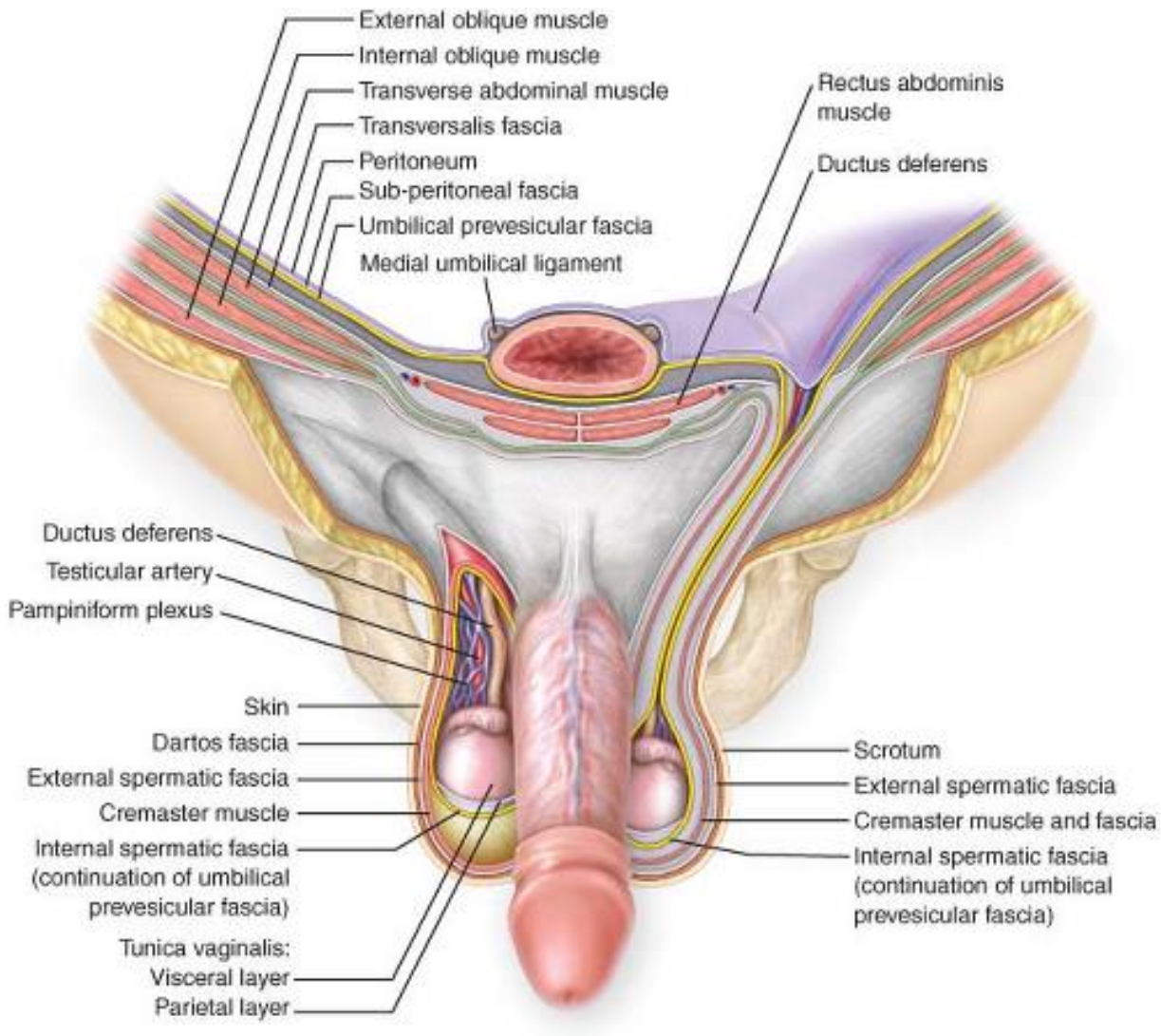
Organs of Sperm Transmission

- Perineal
  - Penis
  - Testes /Epididymis
  - Ductus deferens
  - Bulbourethral gland
- Pelvic
  - Ductus deferens
  - Seminal gland
  - Ejaculatory duct
  - Prostate



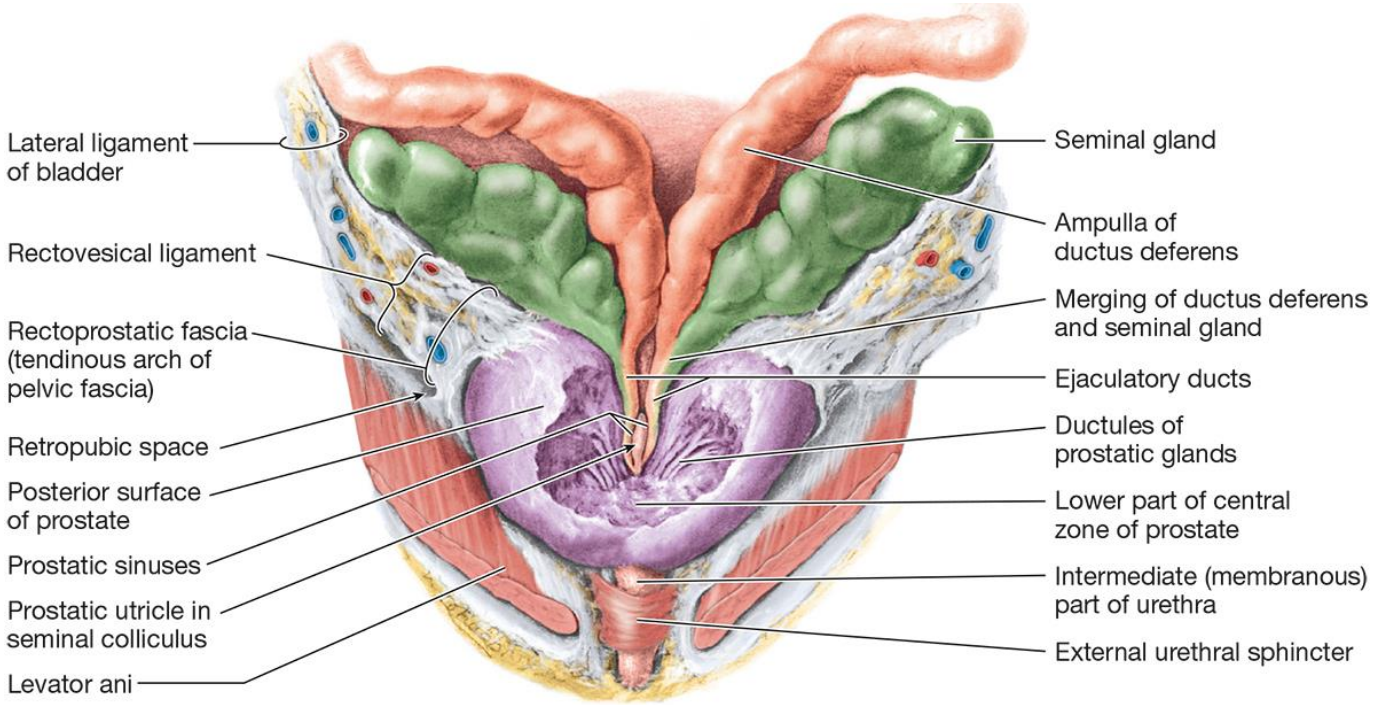
Ductus (Vas) Deferens

- Transmits sperm to the prostatic urethra
- Begins at epididymis
- Runs in the spermatic cord/inguinal canal
- Passes superior to external iliac vessels
- Moves to the posterior aspect of the bladder

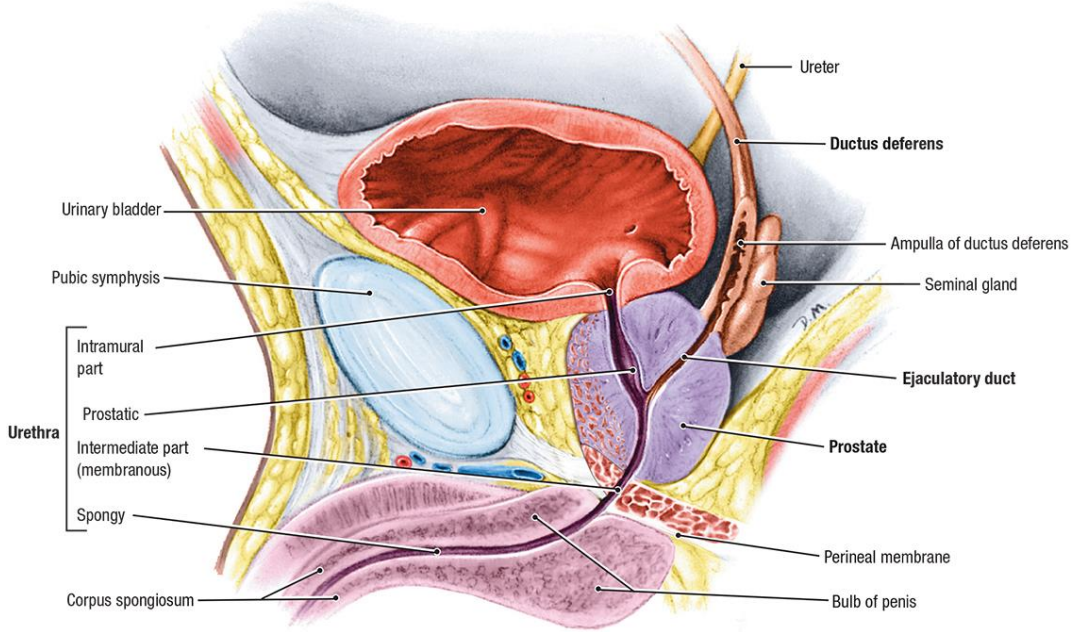


Ductus (Vas) Deferens/ Seminal gland

- Ductus deferens courses over the seminal glands
  - Ends at an ampulla
- Seminal gland is a single long convoluted tube
  - Alkaline secretions
- Vas and seminal gland ducts merge as the ejaculatory duct
  - Pass obliquely through prostate



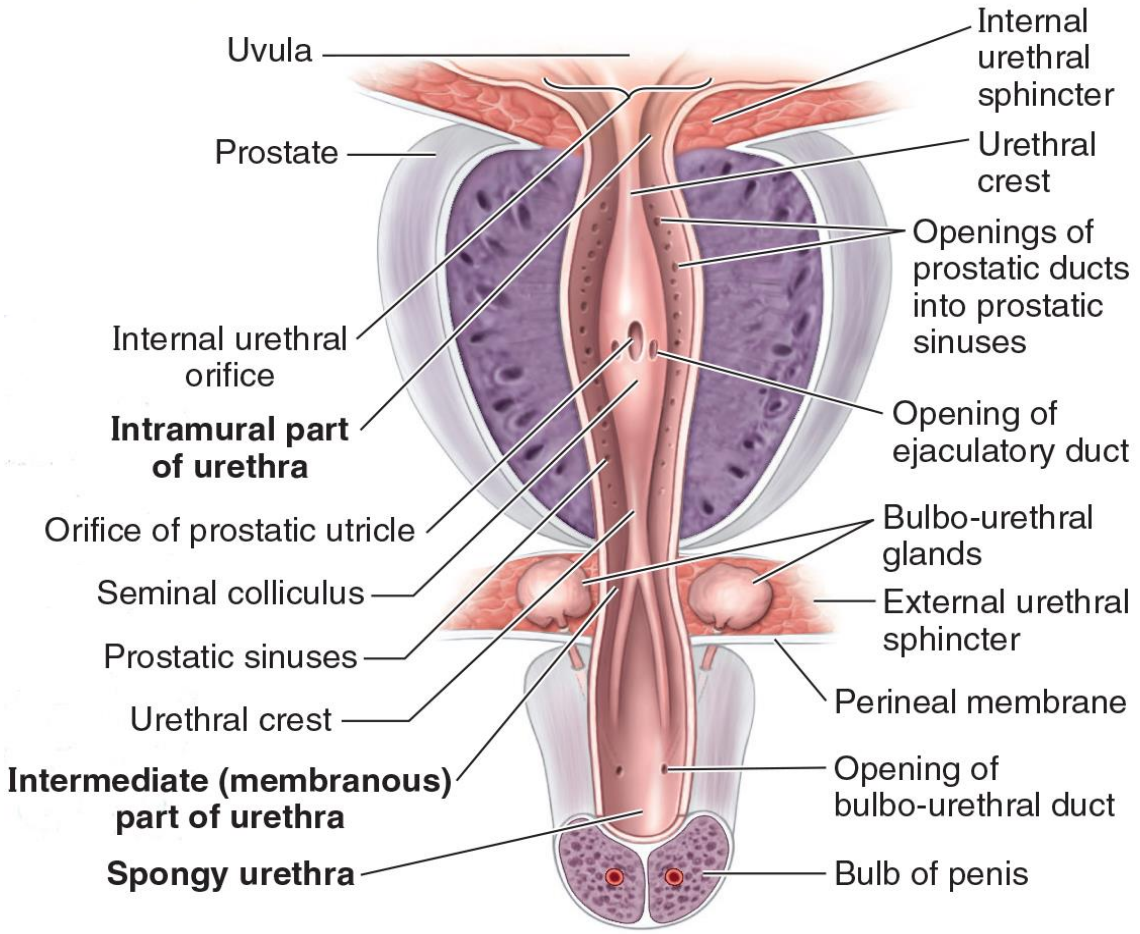
(B) Posterior view



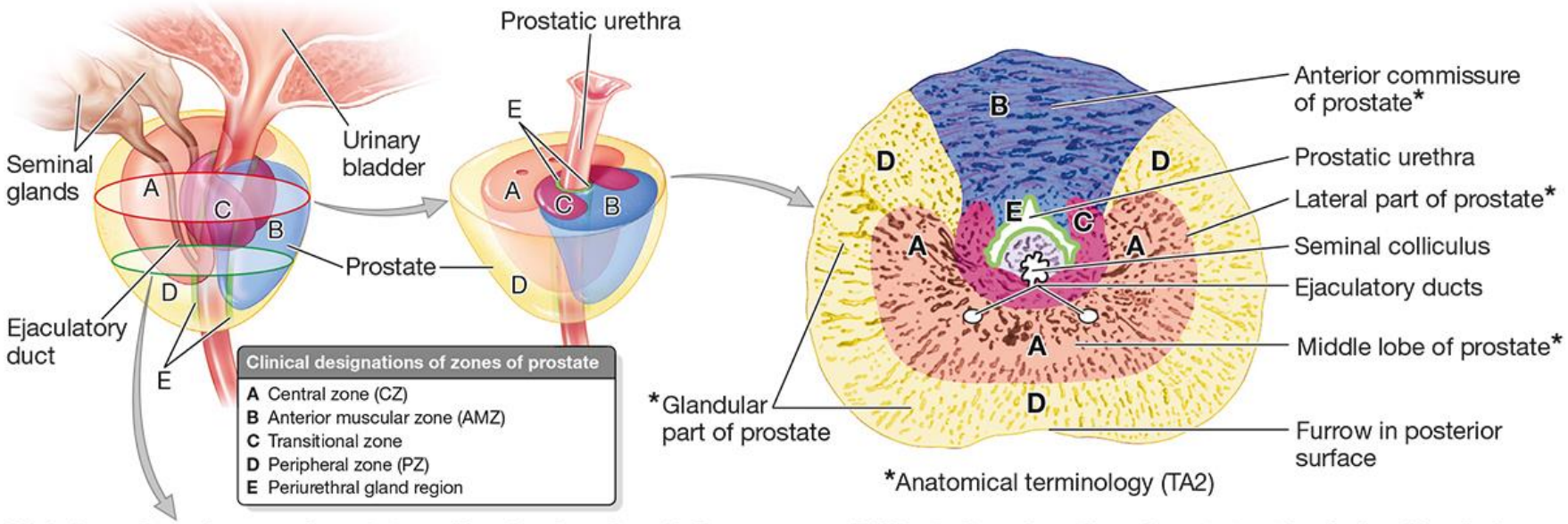
A. Median Section

Prostatic Urethra

- Seminal colliculus on posterior wall of prostatic urethra
- Ejaculatory ducts empty here
- Prostatic utricle
  - Embryonic remnant
  - Can enlarge or get infected



Prostate Gland



(A) Schematics of zones of prostate and levels of sections B–D

(B) Illustration of section of prostate at level of *red line* in A

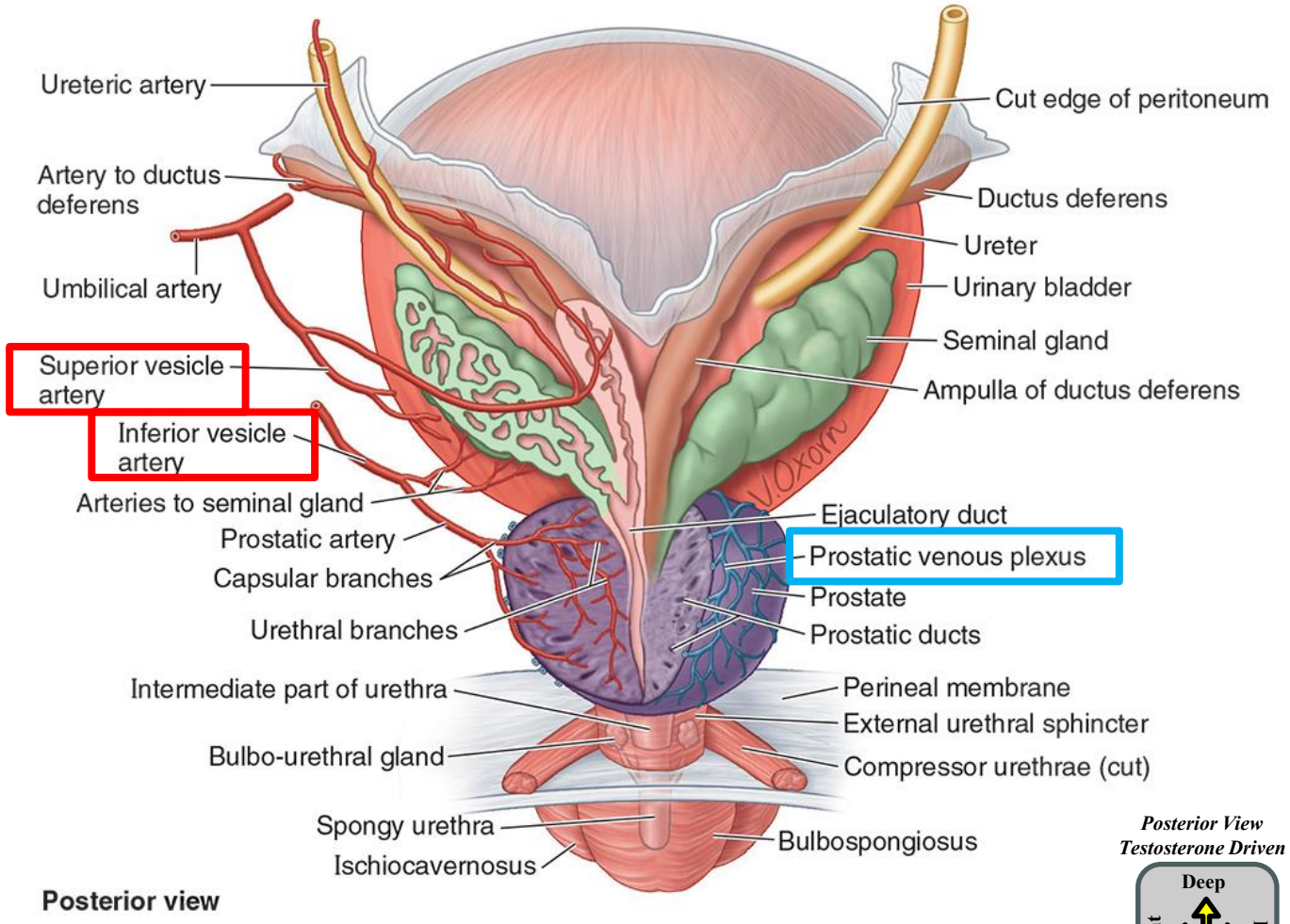
- Encapsulated gland
  - Glandular and fibromuscular portions
- Glandular tissue empties into sinuses adjacent to colliculus
- Several zones
  - Transitional zone – BPH
  - Peripheral zone – most of glandular tissue
    - Digital rectal exam

Vasculature supply of the ductus deferens, ejaculatory ducts and prostate (at the fundus of the bladder)

Arterial supply: branches from the ***superior*** and ***inferior vesicle arteries***

Venous drainage

- Parallels the course and name of the arteries
- ***Prostatic venous plexus*** drains into adjacent plexuses (***vesicle***) or inferior vesical veins
- ***Ductus deferens:*** drains through the distal part of the pampiniform plexus (of the testicular vein) → prostatic plexus

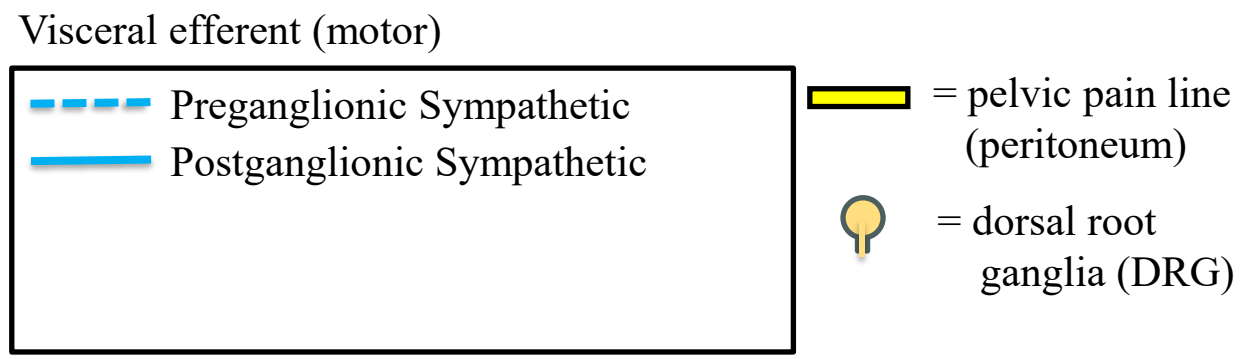


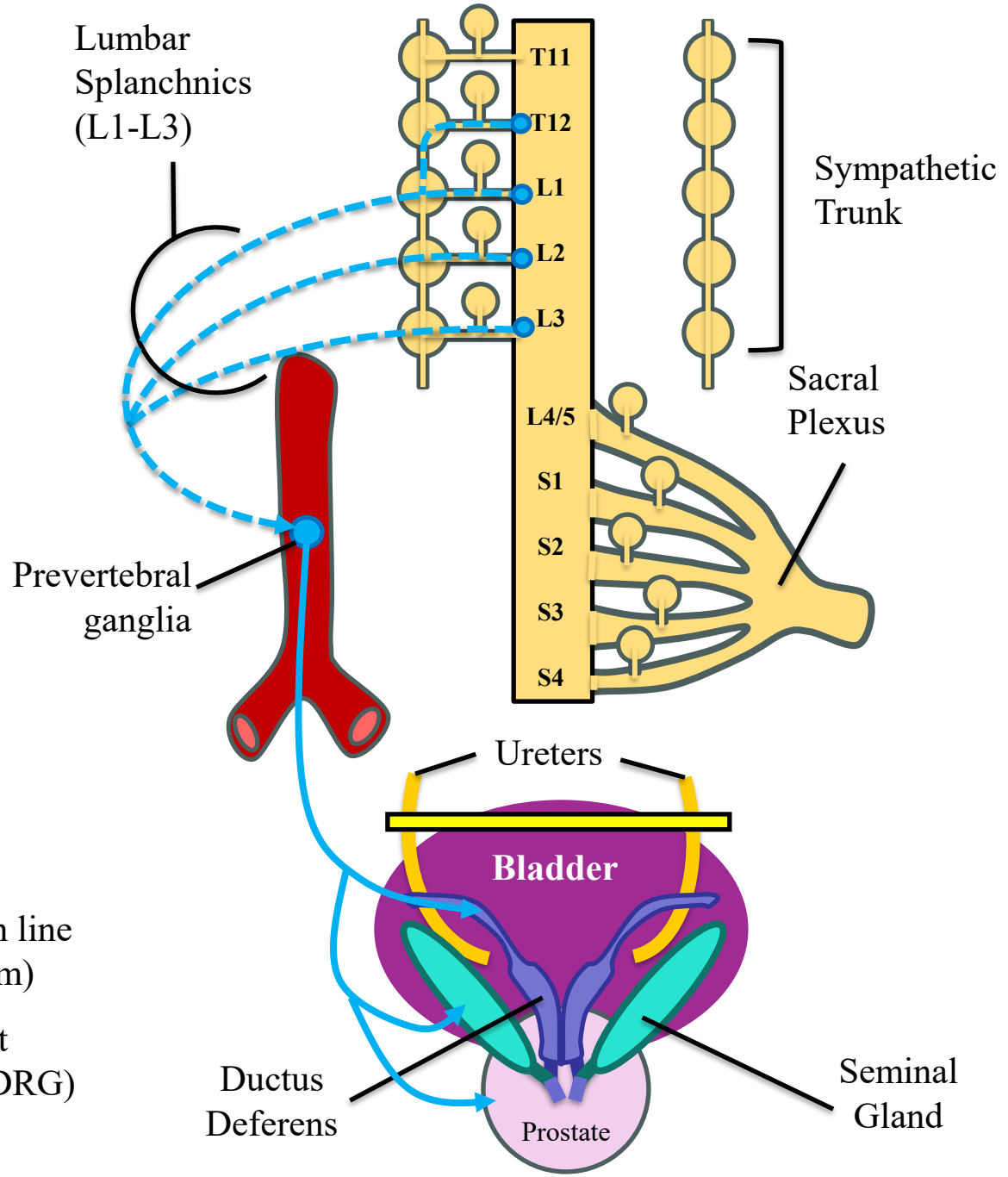
Moore's, 9th ed., Fig. 6.37

Innervation of the ductus deferens, ejaculatory ducts and prostate

Sympathetic

- Spinal levels T12 – L2/3 travel through lumbar splanchnics
- Presynaptic fibers synapse on PREvertebral ganglia (aortic plexus) to reach the target organs via ***pelvic plexuses***
- Contraction of internal urethral sphincter (prevent retrograde of semen), peristalsis of ductus deferens, secretion of seminal and prostate glands

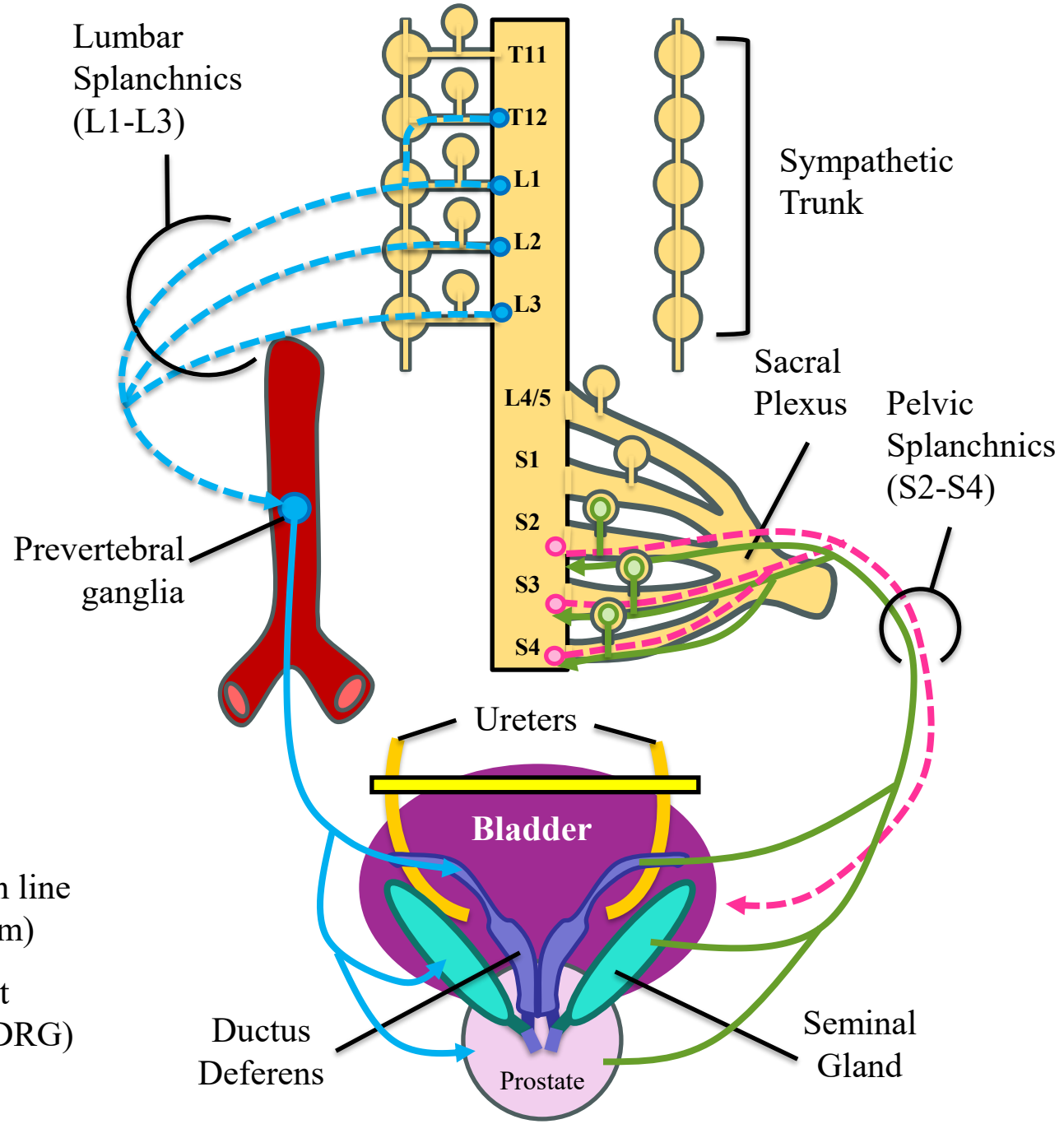




Innervation of the ductus deferens, ejaculatory ducts and prostate

Parasympathetic

- Travel through pelvic splanchnics – penile erection
- ***Visceral sensory*** (***below the pelvic pain line***) return to S2-S4 spinal ganglia



Visceral efferent (motor)

- Preganglionic Sympathetic
- Postganglionic Sympathetic
- Preganglionic Parasympathetic
- Postganglionic Parasympathetic

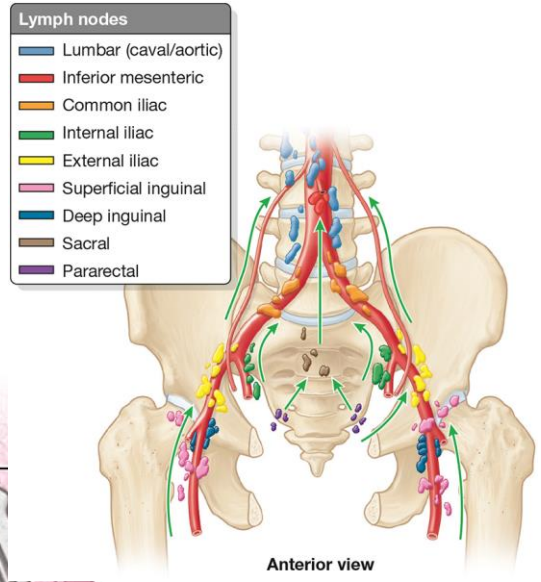
Visceral afferent (sensory)

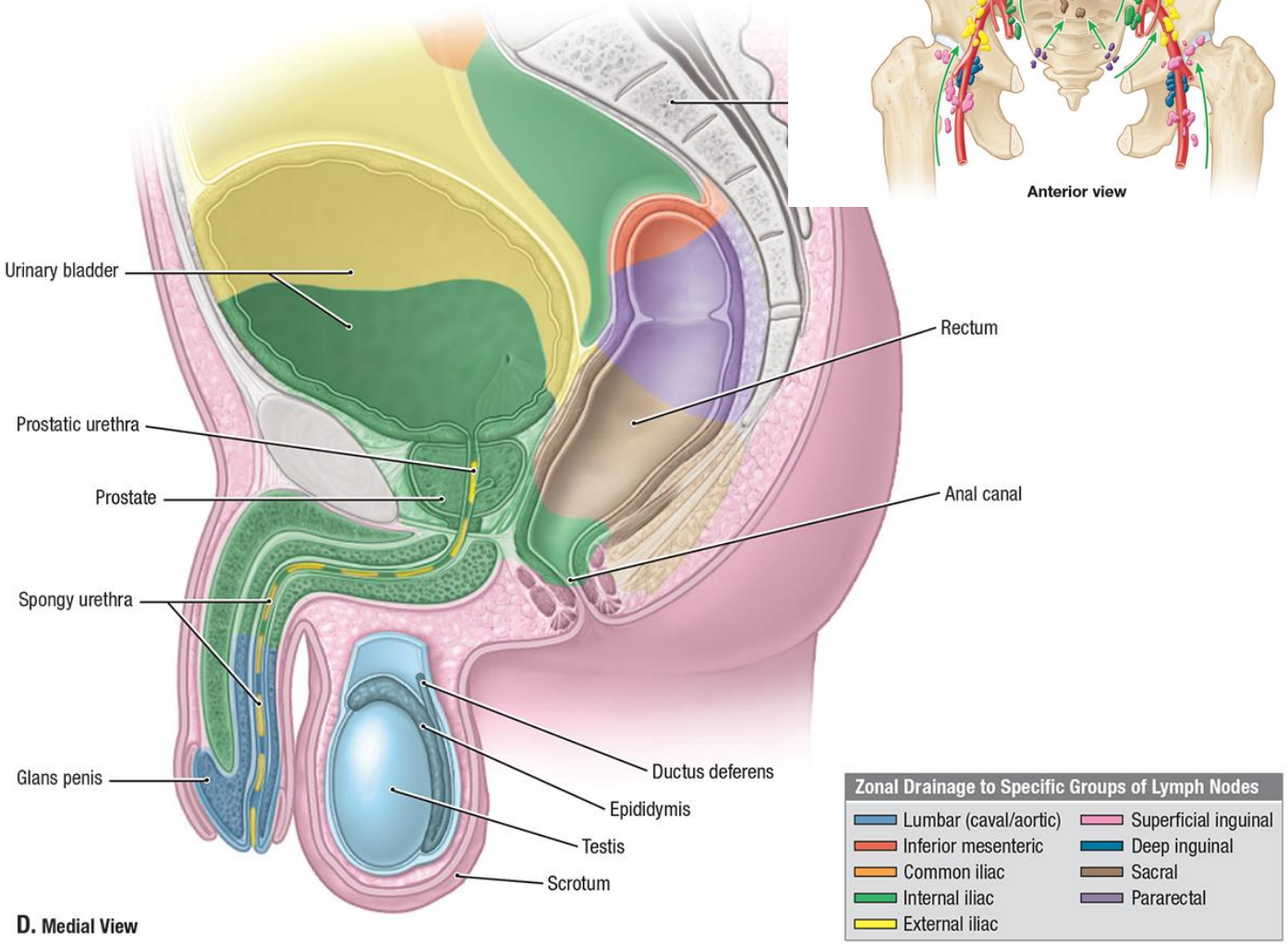
= pelvic pain line (peritoneum)

= dorsal root ganglia (DRG)

Lymphatic Drainage
Organs of Spermatic Pathway

- Testes to Lumbar nodes
  - Follows blood supply
- Internal iliac nodes
  - Prostate
  - Seminal
  - Epididymis
  - Ductus



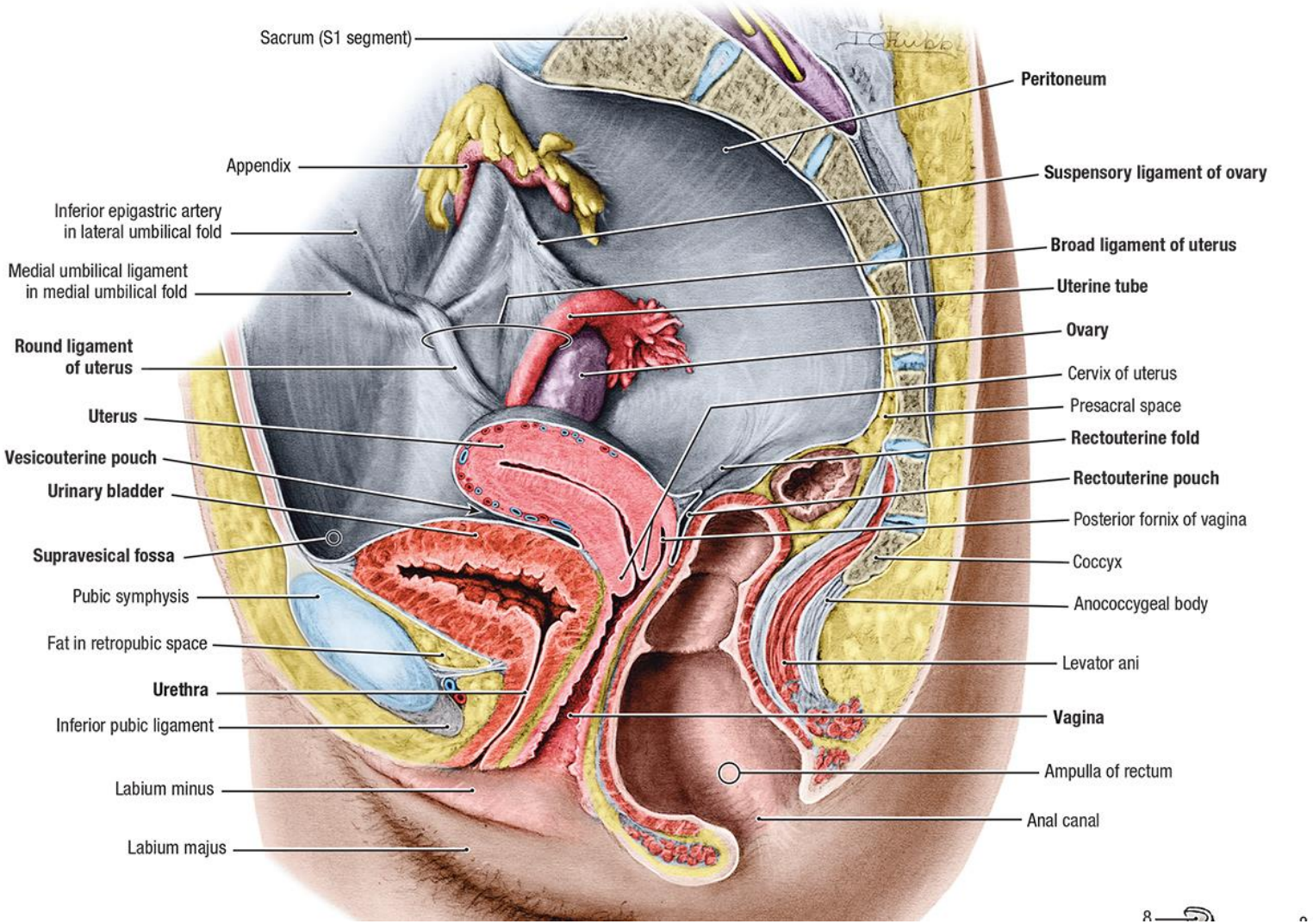


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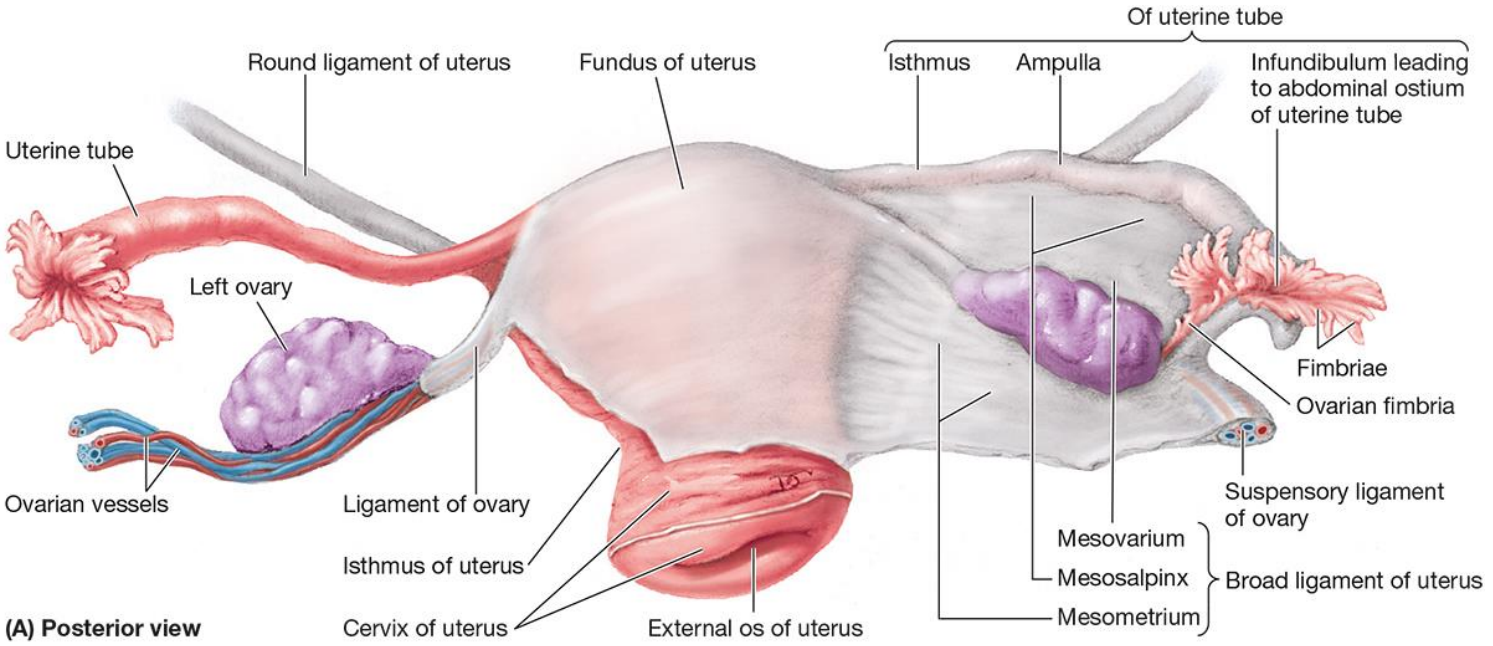
Uterus

- Positioned between urinary bladder and rectum
- Enclosed in peritoneum called the Broad Ligament



Broad Ligament of the Uterus

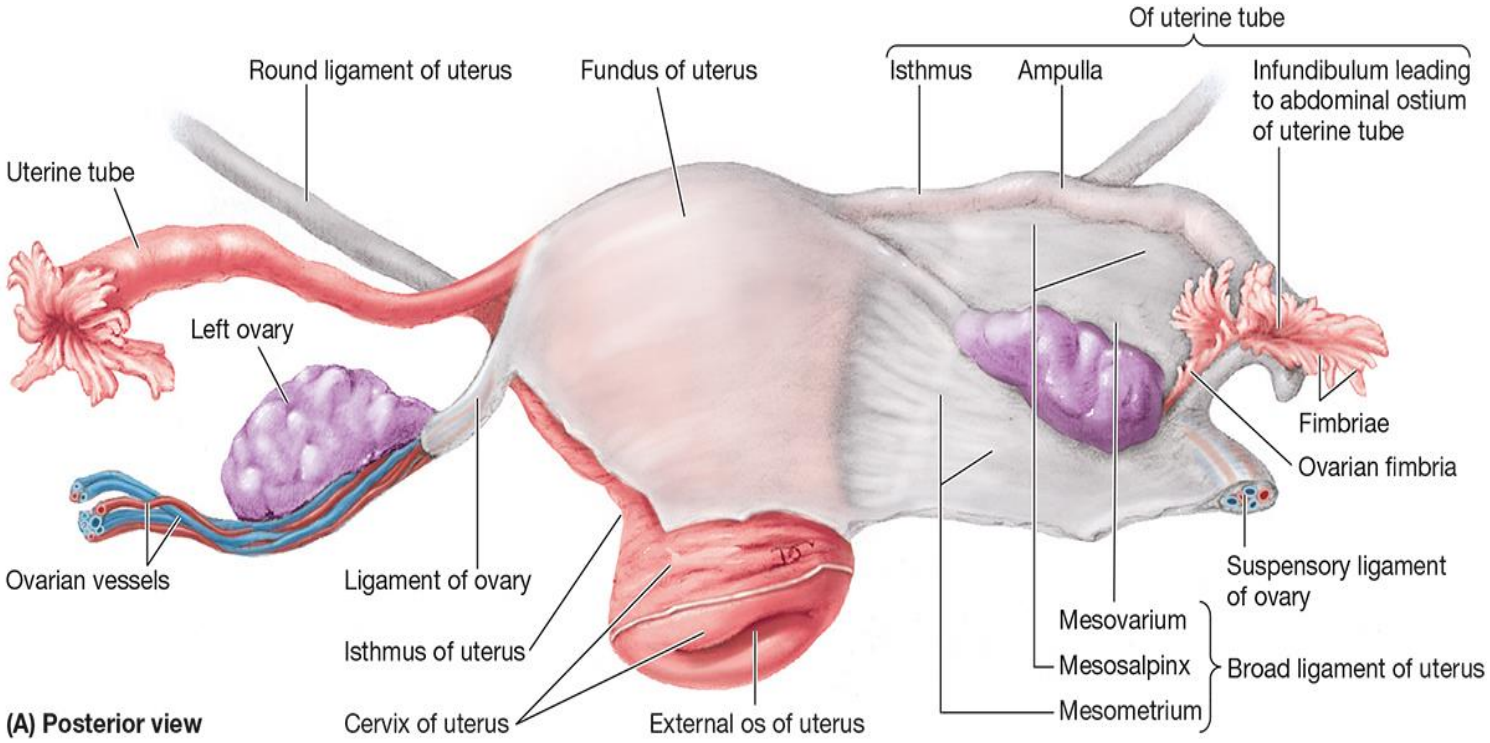
- Peritoneal “poncho”
- Encloses the uterus, uterine tubes, ligament of the ovary, suspensory ligament of ovary
- Does NOT cover ovaries





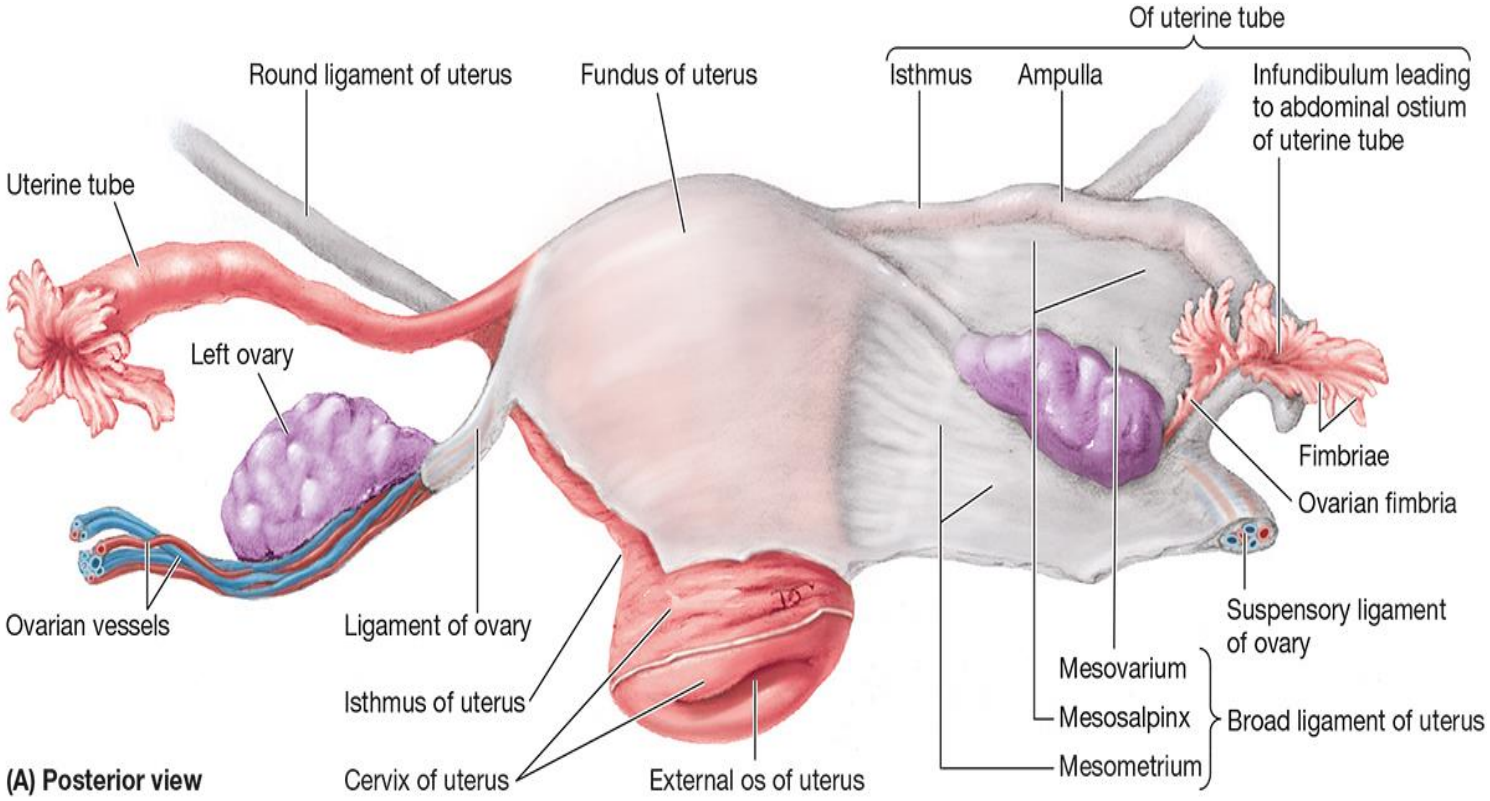
Broad Ligament of the Uterus

- Several regions
  - Mesometrium bulk of ligament
  - Mesovarium part that attaches ovary to ligament
  - Mesosalpinx – part that wraps around uterine tube



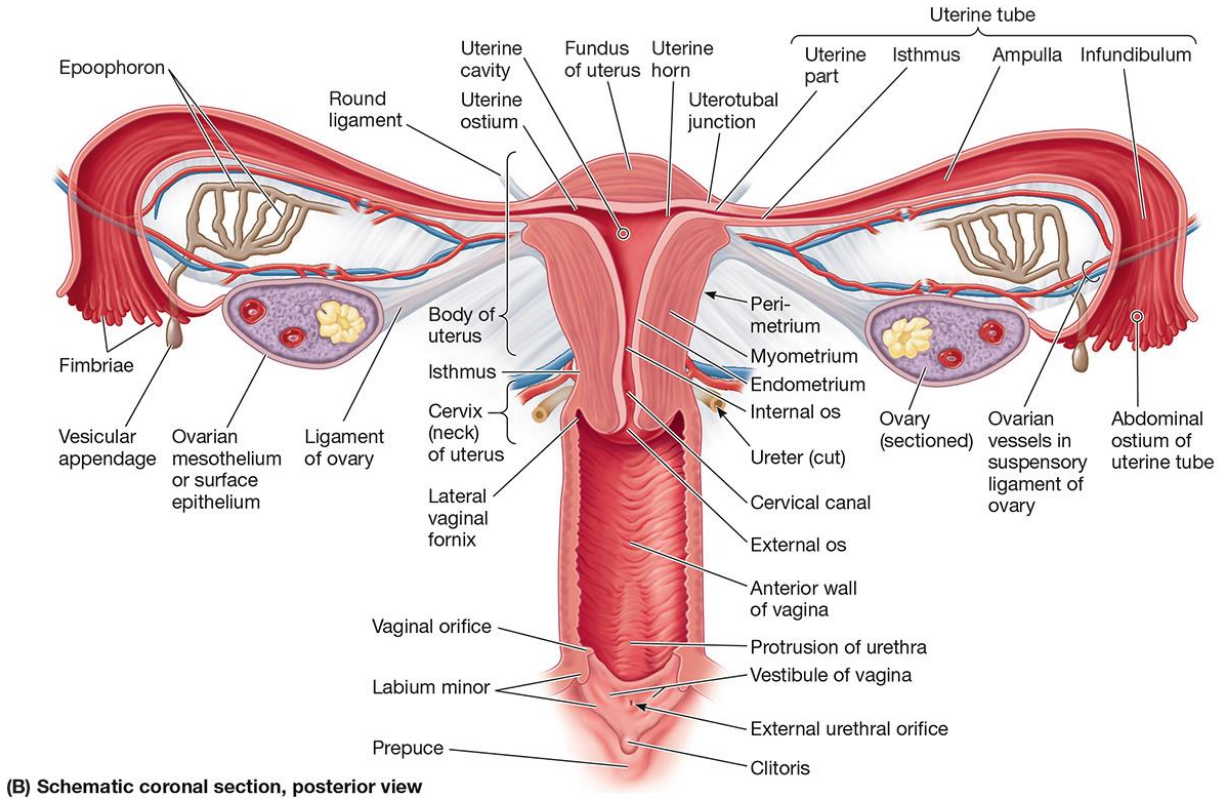
Additional Ligaments of Uterus and Ovaries

- Round ligament
  - Gubernaculum
  - Out the inguinal canal to Labia
- Ligament of the ovary
  - Attaches ovary to the uterine body
  - gubernaculum
- Suspensory ligament of the ovary
  - Carries ovarian vessels and lymphatics



Ovaries

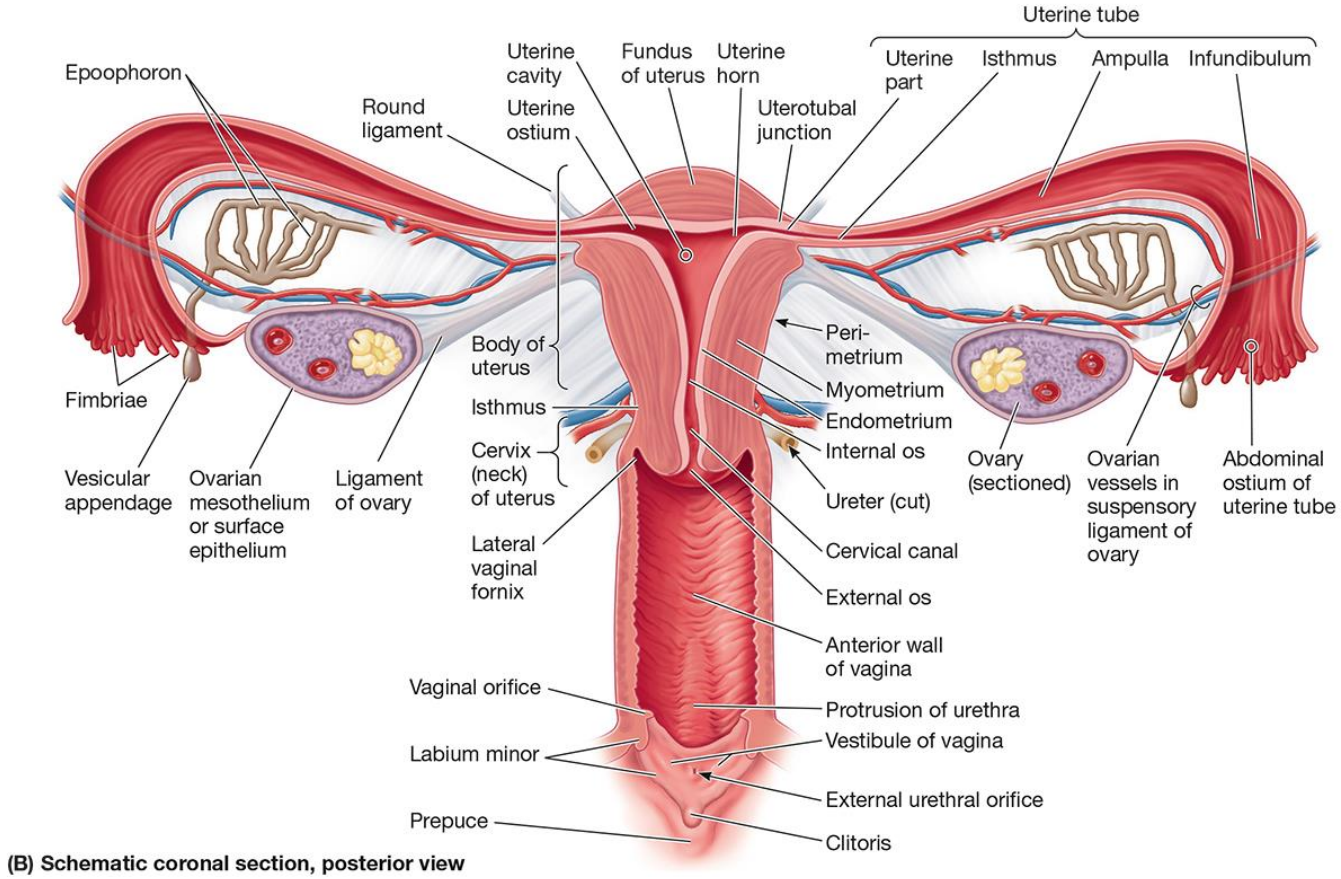
- Gonadal tissue
- Endocrine gland
- Suspended in the peritoneal cavity by the mesovarium
- Oocytes released into the cavity and captured by uterine tubes



(B) Schematic coronal section, posterior view

Uterine Tubes (oviducts, fallopian tubes)

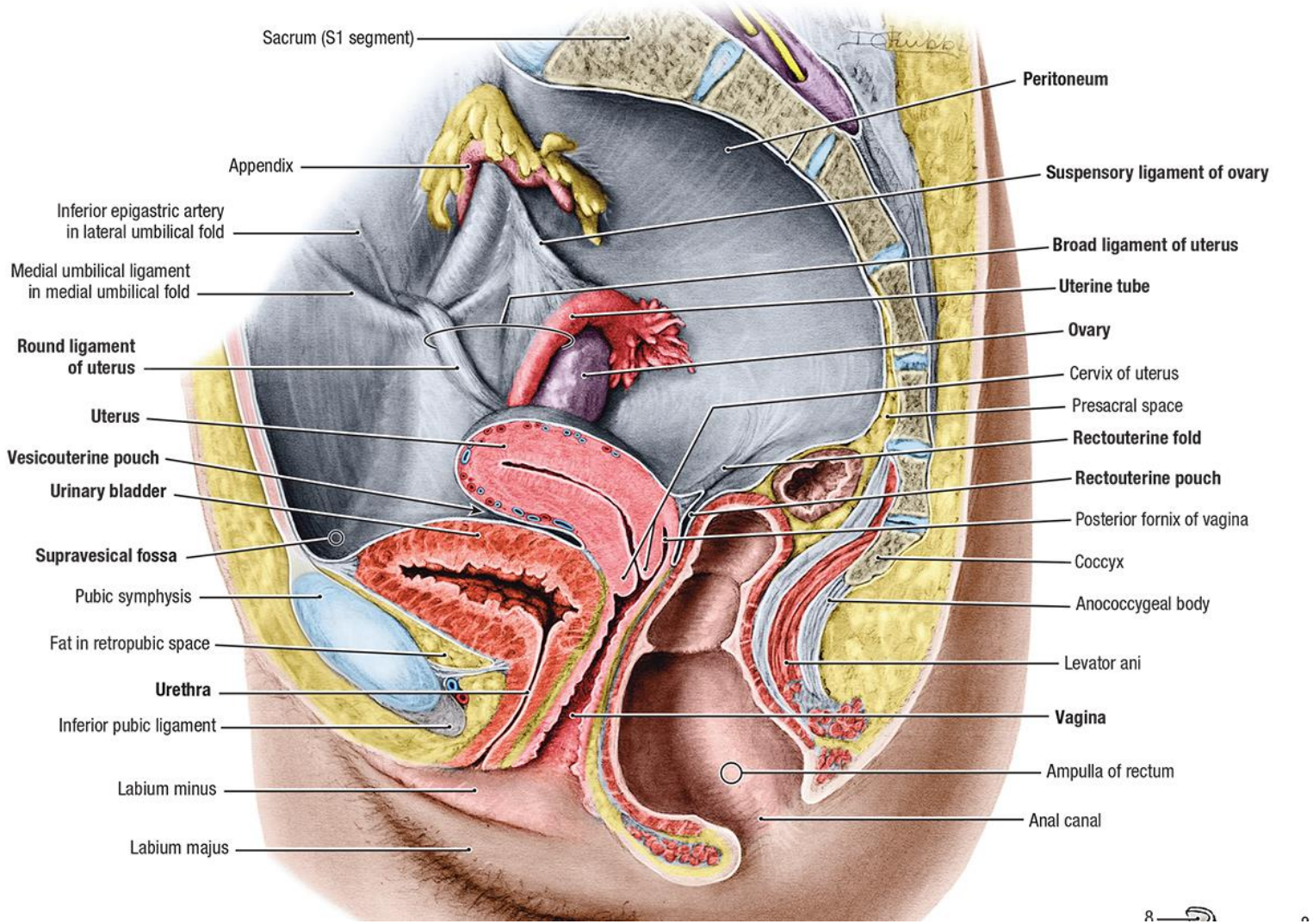
- Covered by the broad ligament
- Open to peritoneal cavity
- 4 parts
  - Infundibulum
    - fimbriae
  - Ampulla
    - Site of fertilization
    - Ectopic pregnancy
  - Isthmus
  - Uterine and uterine ostium



(B) Schematic coronal section, posterior view

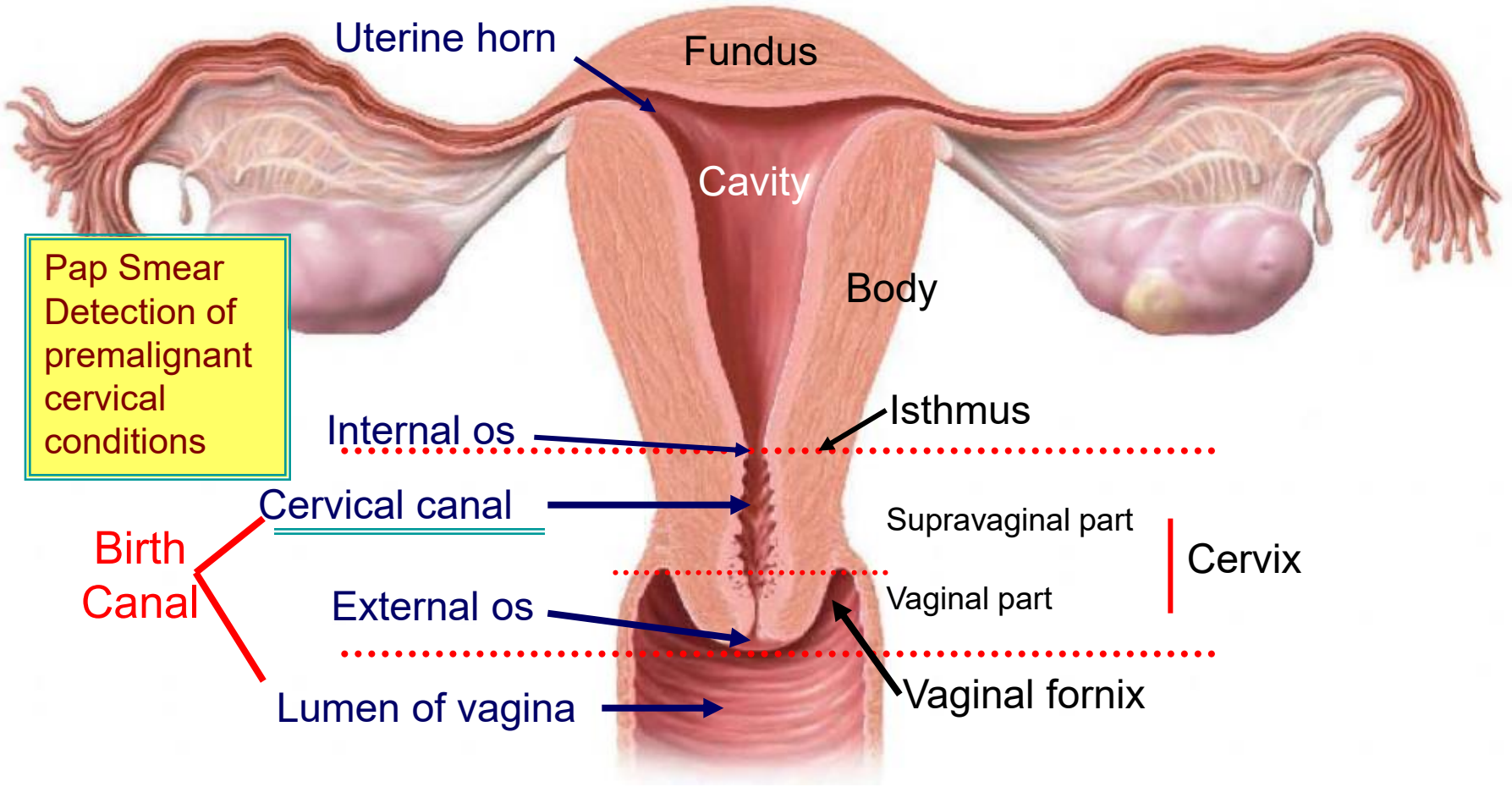
Uterus

- Thick muscular organ
- Cervix projects into the vaginal cavity
- Draped forward over the bladder
  - Anteverted cervix
  - Anteflexed body



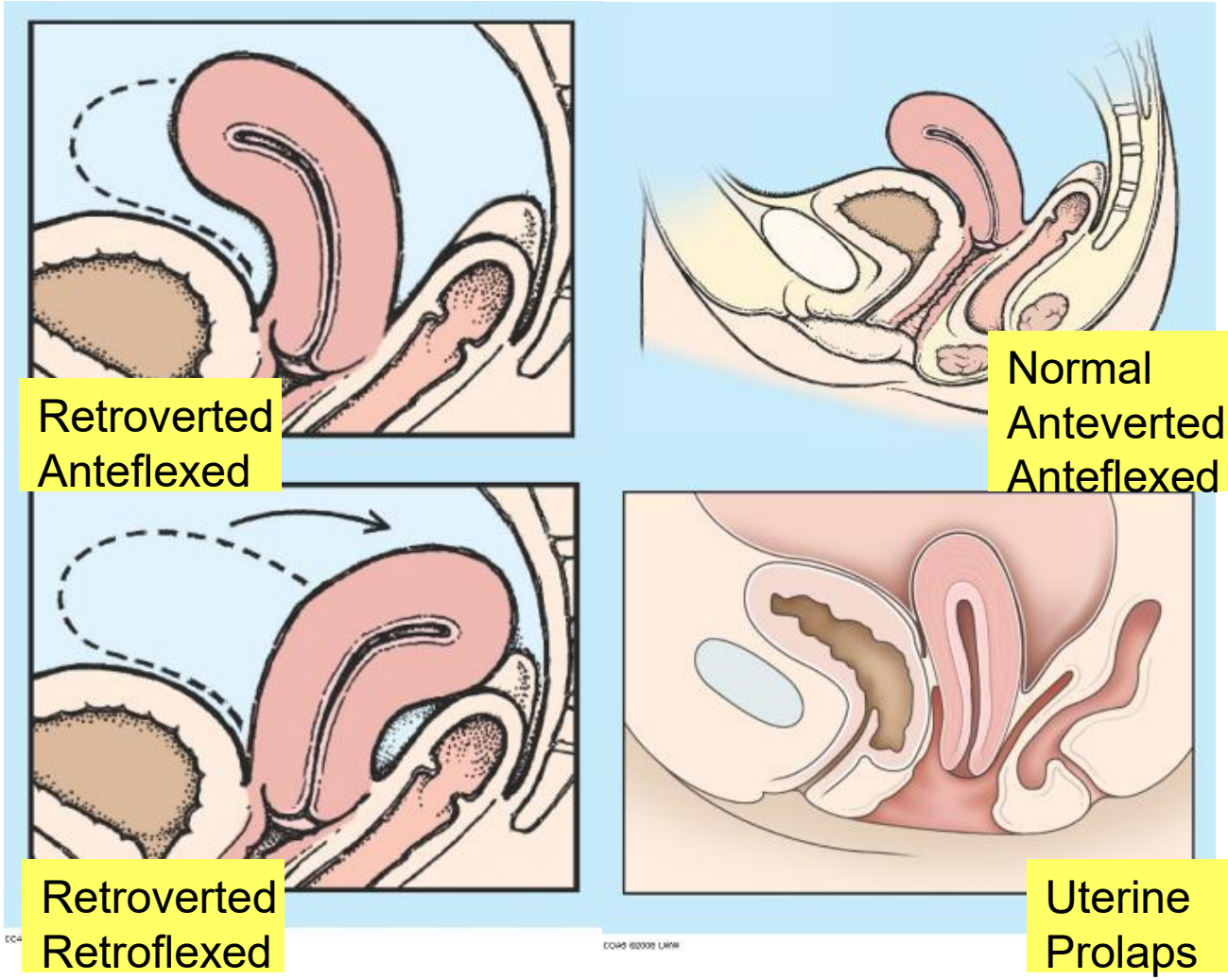
Uterus

- Fundus
- Body
- Isthmus
- Uterine cavity
  - Uterine horn
- Cervical canal
  - Internal and external os



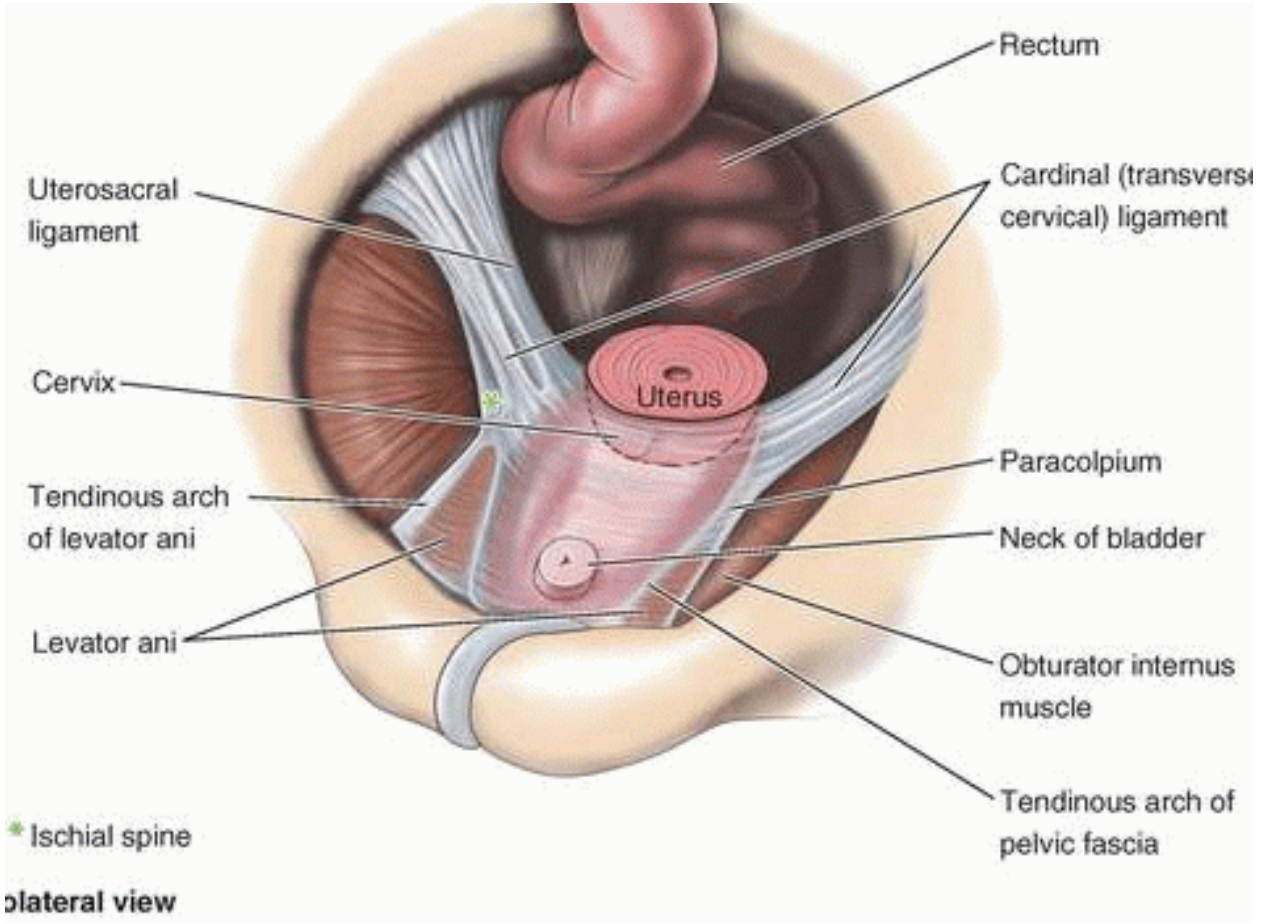
Changes in Uterine Positions

- Changes in uterine position can affect pregnancy, delivery or prolapse
- Retroflexed
  - Uterine body is bent backwards
- Retroverted
  - Cervical canal positioned more vertical

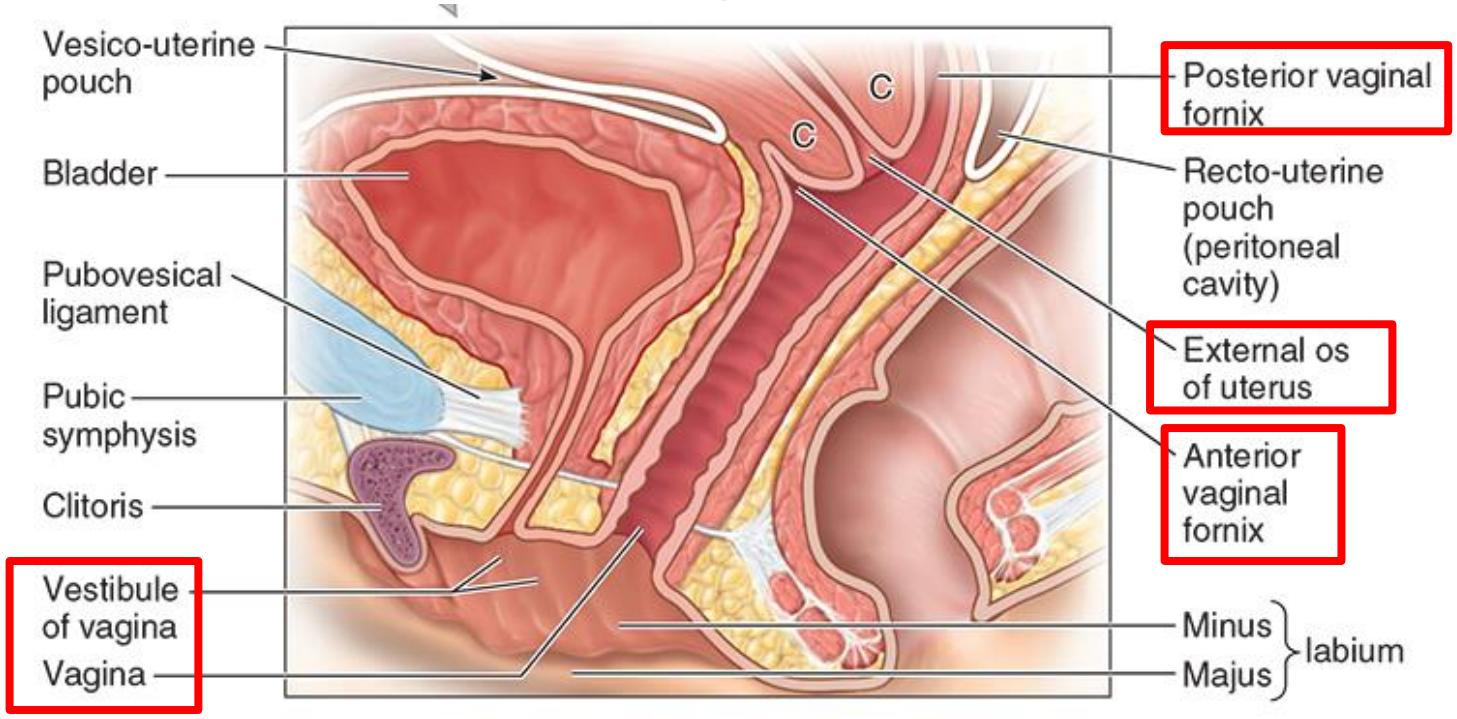


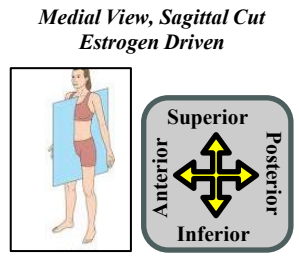
Uterine Supports

- Active/Dynamic support from the pelvic diaphragm
  - Kegels to strengthen
- Passive support
  - Uterine ligament are thickened fascia
  - Cardinal ligaments/transverse cervical
  - Uterosacral



Vagina





- Distensible musculomembranous tube
- From superior most vaginal part of the cervix of the uterus to the vaginal orifice
- ***Vaginal orifice*** opens into the vestibule of the vagina (between labia minora)
- Posterior to the fundus of the bladder; medial to levator ani; anterior to anal canal, rectum and recto-uterine pouch
- The ***vaginal fornix*** (narrow recess) surrounds the vaginal part (***external os***) of the cervix

Moore's, 9th ed., Fig. 6.43

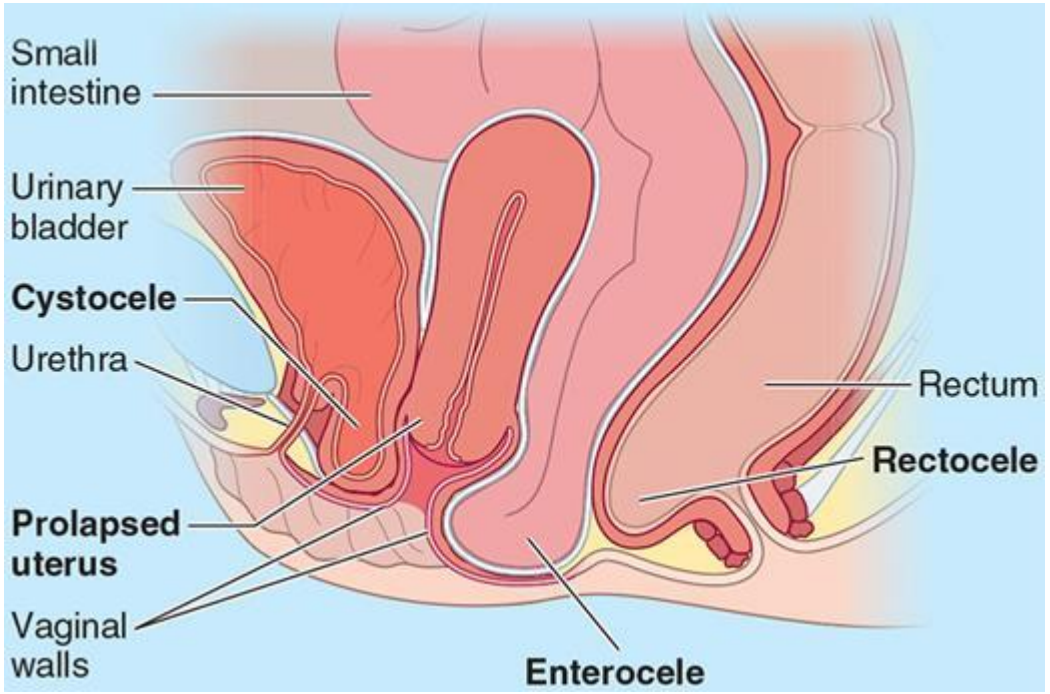
Organ Prolapse

Cystocele: prolapse of the anterior vaginal wall involving the bladder

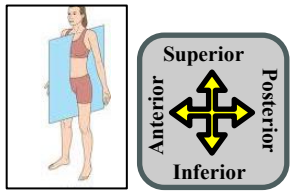
Uterovaginal prolapse: prolapse of the uterus, cervix, or upper vagina

Rectocele: prolapse of the lower posterior vaginal wall involving the rectum

Enterocele: prolapse of the upper posterior vaginal wall involving the rectovaginal pouch

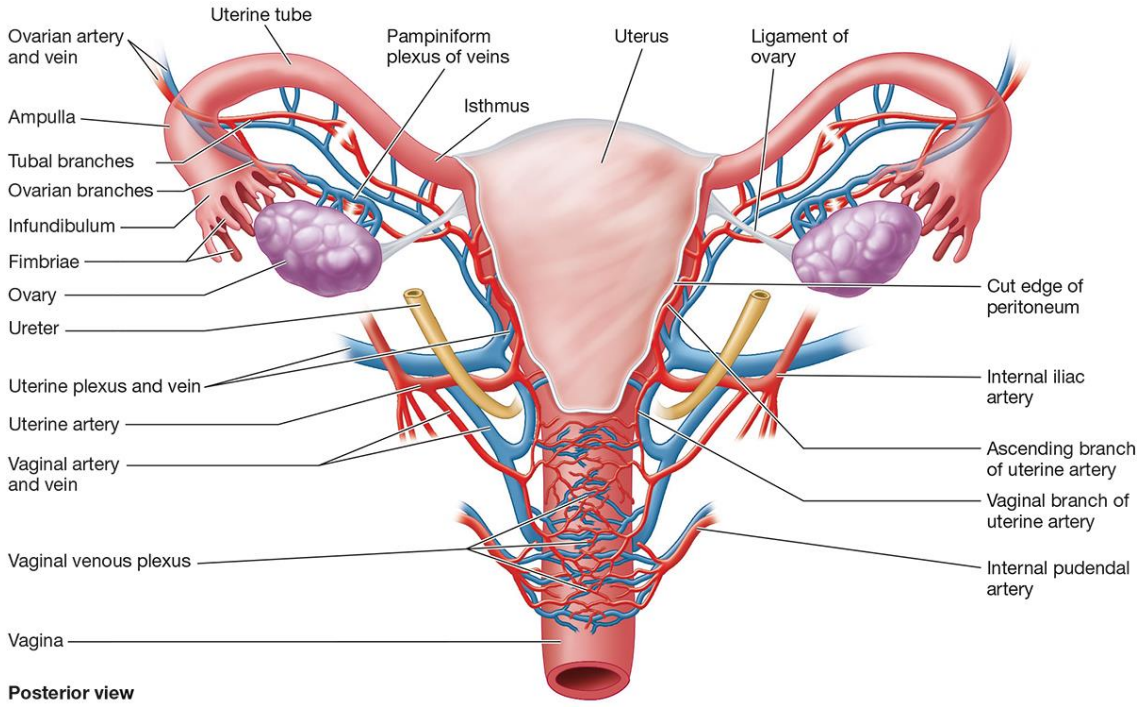


Medial View, Sagittal Cut
Estrogen Driven



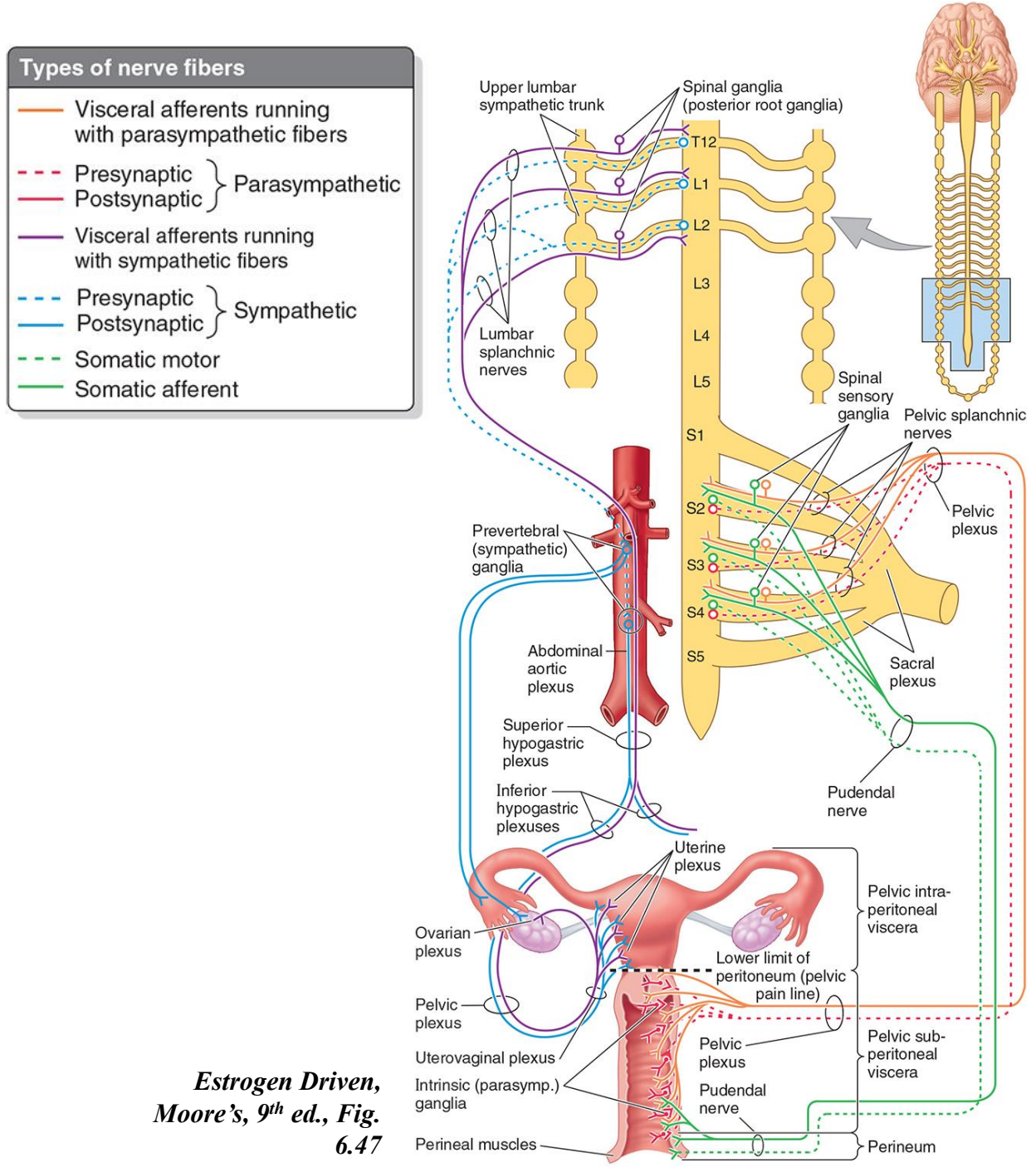
Blood Supply

- Ovarian arteries from aorta
  - In suspensory ligament
- Uterine artery from internal iliac
  - Ureter passes under
  - Anastomoses with ovarian along fallopian tubes
  - Changes during ovulation
- Vaginal arteries from uterine
- Internal pudendal vagina in perineum
- Ovarian veins drain to IVC/renal vein
- Uterine and vaginal venous plexuses



Autonomic Innervation of the vagina and uterus

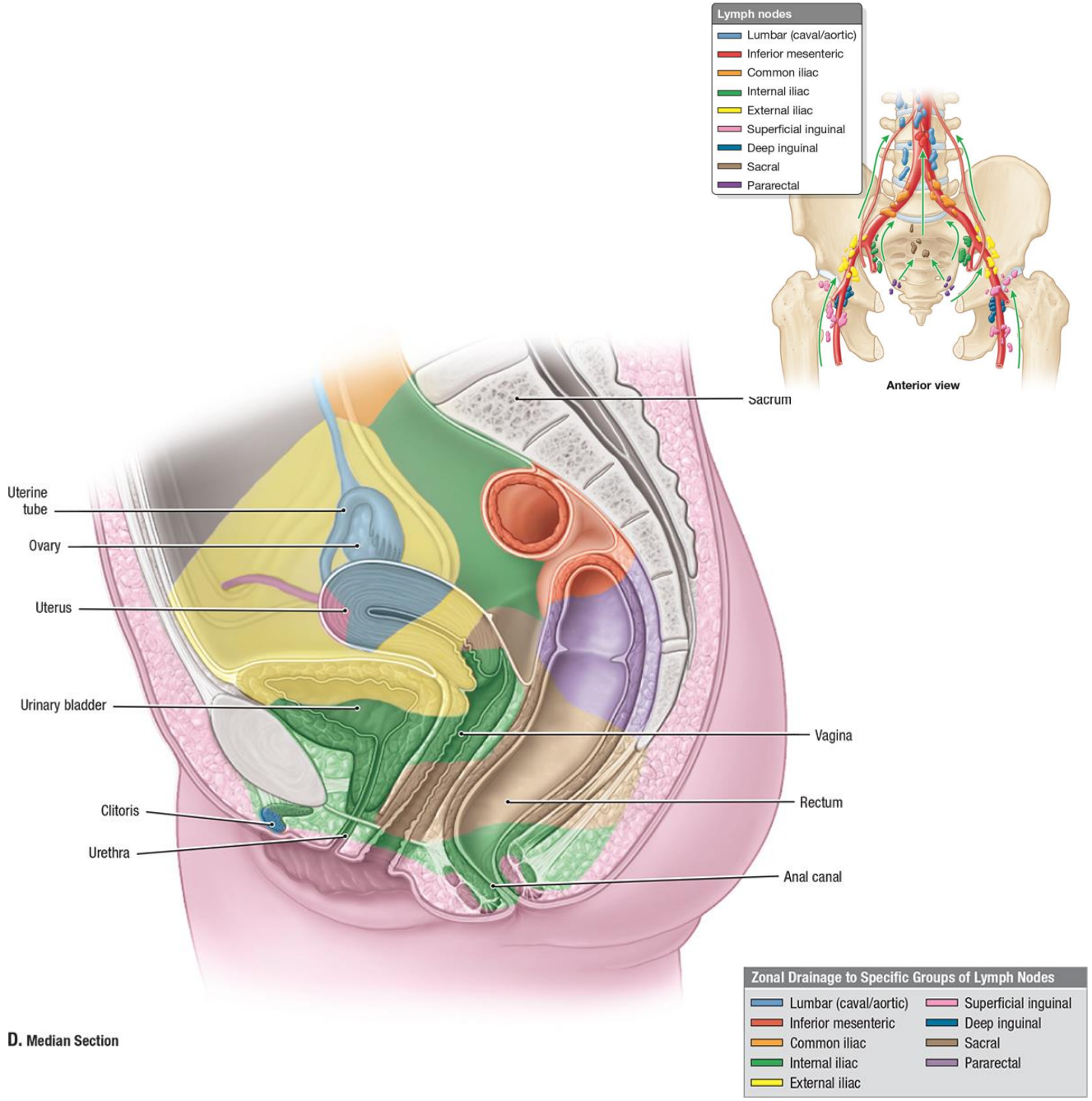
- ***Sympathetic*** (visceral motor): T12-L2/3 spinal levels via lumbar splanchnics through the ***hypogastric plexuses***
- ***Parasympathetics*** (visceral motor) – ***pelvic splanchnics*** (S2-S4) to ***inferior hypogastric plexus***



Estrogen Driven, Moore's, 9th ed., Fig. 6.47

Lymphatic Drainage

- Round ligament to superficial inguinal
- Ovaries and distal fallopian tubes to lumbar
- Vagina to internal iliac and sacral
  - Distal to superficial inguinal
- Cervix to superficial inguinal and internal iliac
- Uterus multiple drainage sites



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Afferents of Estrogen Driven Pelvic Viscera

Somatic sensory (S2-S4): inferior ¼ of vagina – touch temperature and pain via deep perineal branch of pudendal

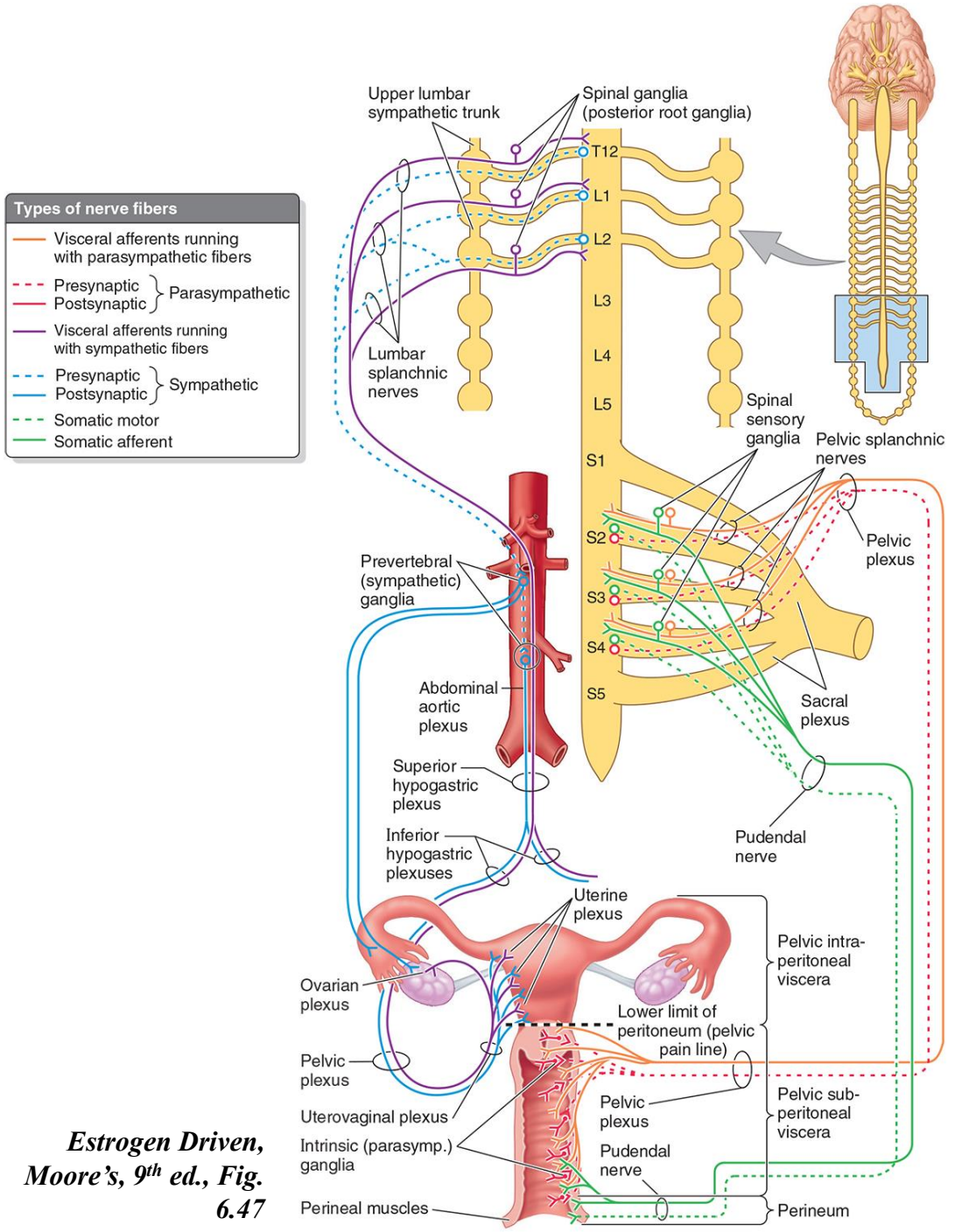
Visceral Sensory

Vagina and cervix (***inferior to the pelvic pain line***)

- ***Parasympathetics*** - travel through pelvic splanchnics to reach spinal ganglia at S2-S4

Uterus (except cervix), uterine tubes, and ovaries (***superior to the pelvic pain line***)

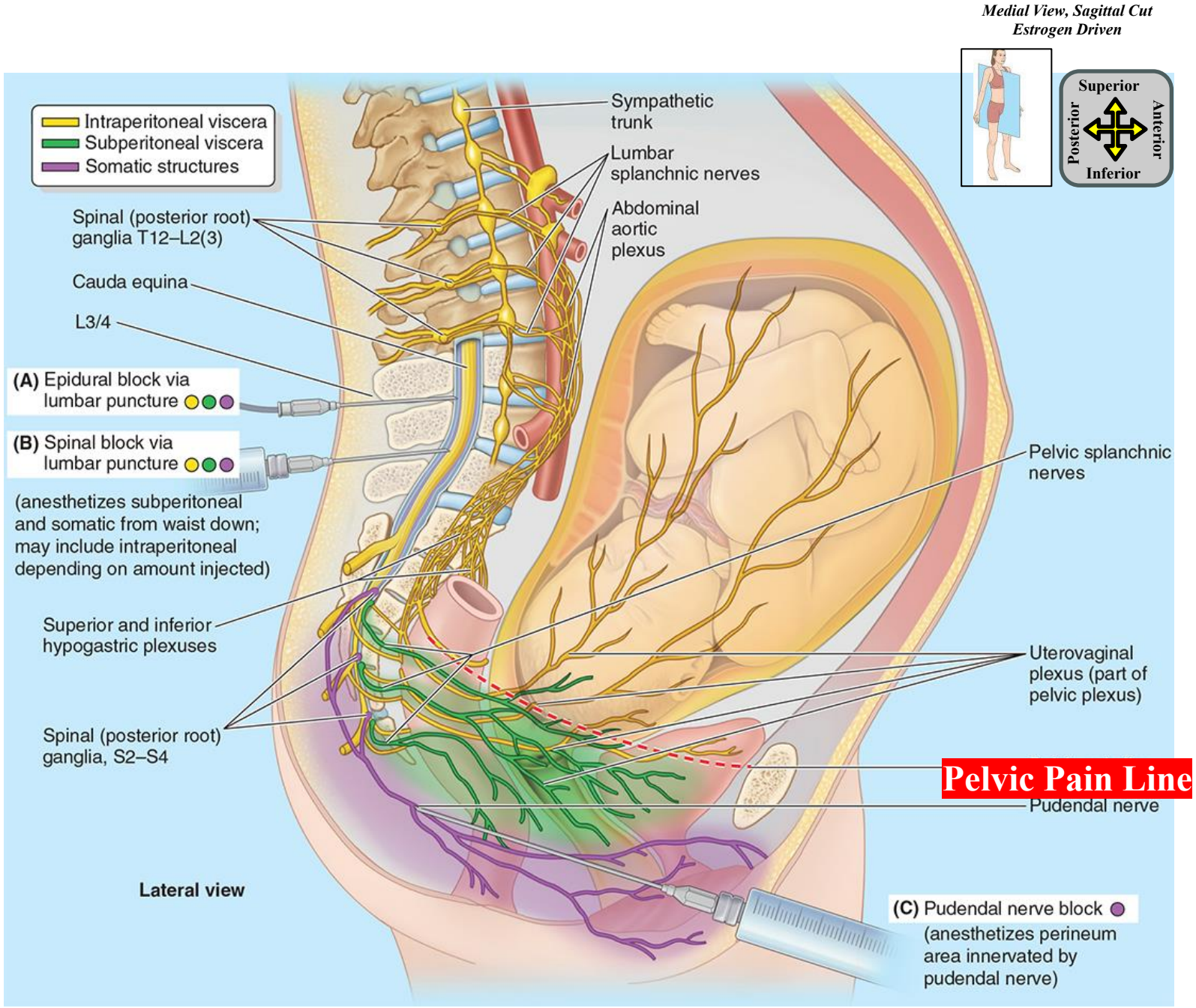
- ***Sympathetics*** - travel through lumbar and sacral splanchnics to reach inferior thoracic spinal ganglia at T12-L2



Estrogen Driven, Moore's, 9th ed., Fig. 6.47

Anesthesia for Childbirth

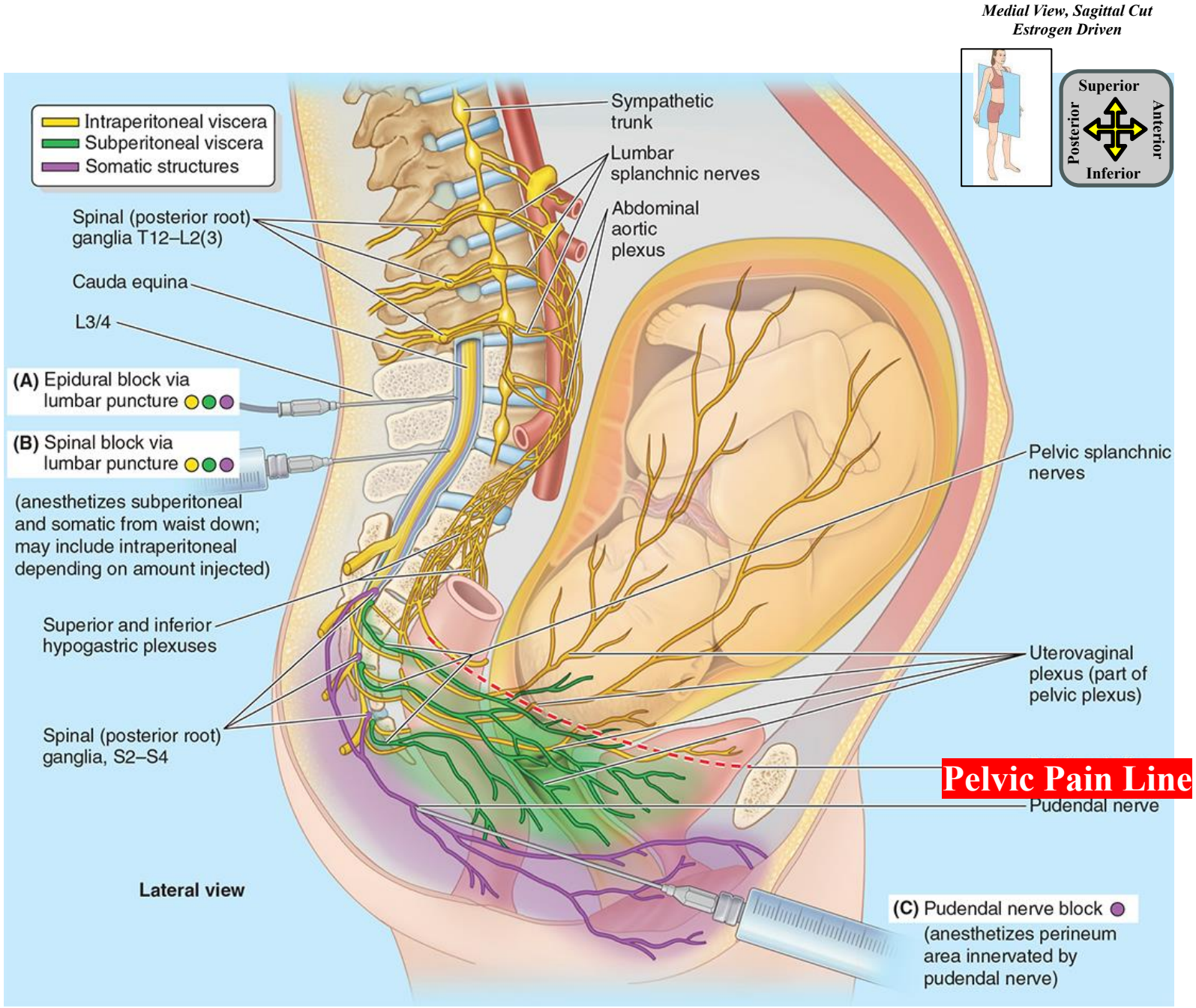
- Epidural
  - Epidural space L3/4
  - Catheter preplanned longer administration
  - Numbs birth canal and perineum
  - Not legs or uterine body
  - Does not ascend to brain



Moore's, 9th ed., Fig. B6.27

Anesthesia for Childbirth

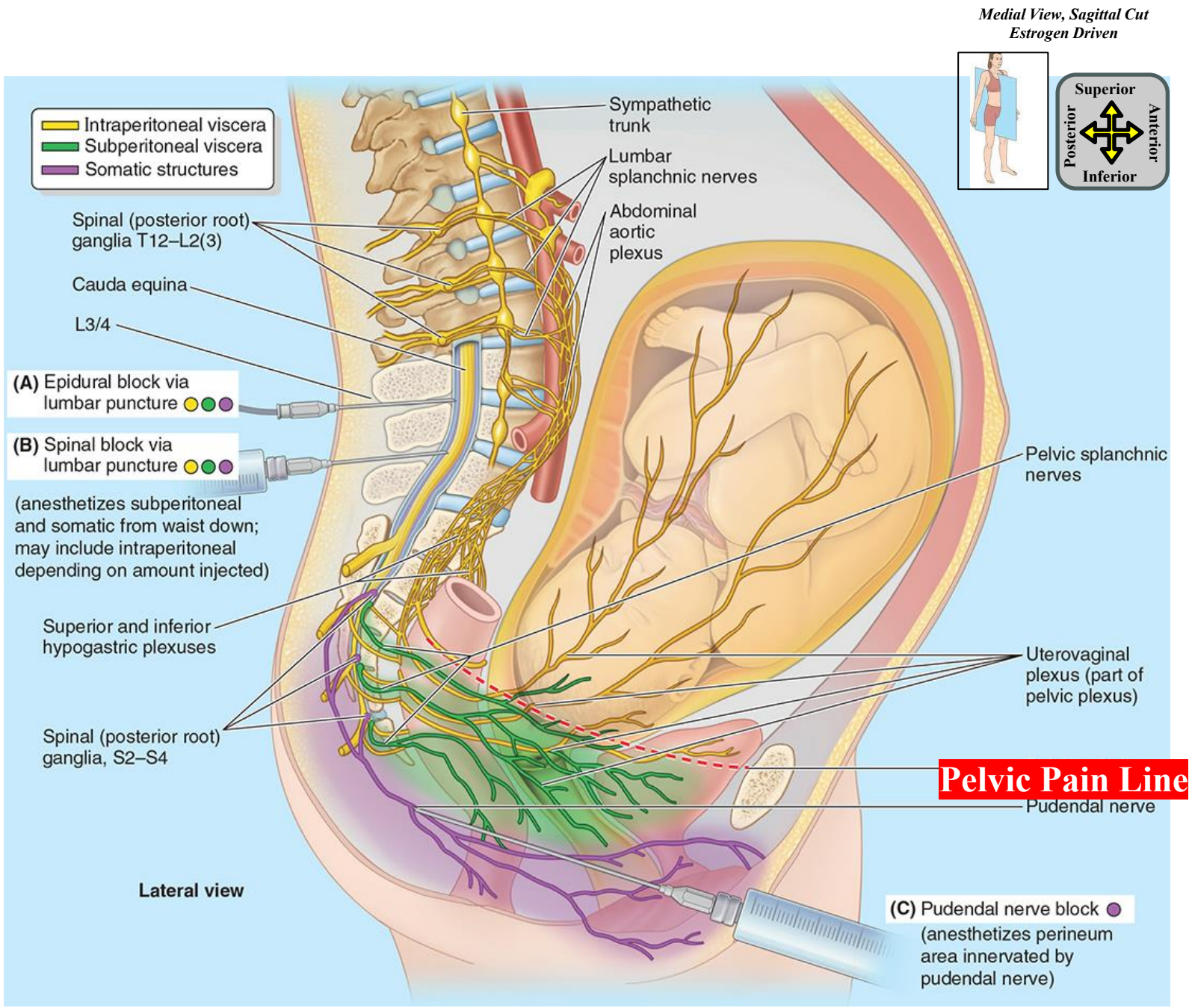
- Spinal
  - Into subdural space
  - L3/4 or L4/5
  - Everything from waist down
  - Short lasting
  - Typically stays in spinal area
  - Can circulate to cerebral spaces and yield headache



Moore's, 9th ed., Fig. B6.27

Anesthesia for Childbirth

- Pudendal
  - Peripheral nerve block at the perineum
  - Numbs the distal vagina and perineum
  - Able to feel the contractions
  - Can be readministered if needed
  - Ischial spine used as landmark



Moore's, 9th ed., Fig. B6.27